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KINEMATIC ANALYSIS AND SIMULATION OF THEO JANSEN MECHANISM

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ABSTRACT

Legged robots have come to researchers' attention due to their abilities to surmount or deal with complex obstacles, high levels of mobility (changing direction while moving), active suspension using force-controllable actuators in confronting irregular terrains, maneuverability and terrain adaptability. Within a few decades, Theo Jansen's wind-powered beach animals found their place among the other types of legged machines. These autonomous creatures are able to live by their own, storing wind energy to move their PVC-made legs, detecting wet sands by the help of implanted nose feelers and escaping from danger by anchoring themselves to the ground. The advantage of these animals is that the required power of the engine must be sufficient to transfer only the legs, and not the transported elements. In this paper we employed the loop closure equations method using MATLAB for kinematic analysis of a Theo Jansen mechanism. Moreover, we performed simulations in SolidWorks to validate the method.

INTRODUCTION

Lately, there has been much work done on the synthesis, analysis and optimization of Theo Jansen mechanisms using programming software including Mathematica, MaTX and Borland's Delphi. Ghassaei [1] uses Mathematica for the design of the Theo Jansen mechanism using the Chebyshev replication and an approximation approach, where the latter simulates the links' movement as sine and cosine functions in order to produce desired loci, and performs an optimization to extract the optimal links length. However, no systematic approach for the kinematic analysis is provided in [1]. Nansi. *et al.* [2] implements a comprehensive dynamic analysis based on the projection method using the MaTX, which is a continuation of the proposed work by Blajer [3]. Ingram [4] examines a genetic algorithm to optimize foot trajectory. Ingram [5] also employs an elementary numerical algorithm for kinematic as well as kinetic analysis using Borland's Delphi software.

In this work, we present a complete, straightforward, yet precise kinematic analysis of the Theo Jansen mechanism based on the loop closure equations method, using MATLAB to provide the preliminary kinematic data as a basis for future examinations of this novel walking mechanism such as dynamic analysis and optimization. To validate the analytical data obtained in MATLAB, we performed simulations of the paired legs using the SolidWorks Motion Study package.

DESCRIPTION OF THEO JANSEN MECHANISM

Figure 1 shows the Theo Jansen mechanism, which is a single degree of freedom mechanism composed of eight or more legs. Applications of this marvelous legged mechanism go beyond human-powered machines such as multi terrain personal transport [6], multi terrain wheel chair, beach vendor carts, robotic house pets, and steampunk walking ship [7].

Figure 2 depicts a schematic of one leg of the Theo Jansen mechanism which has a crank (m), two oscillating rockers (b,c), and two couplers (j,k), all connected by pivot joints. This linkage is very similar to a Klann linkage [8]. Each leg consists of six parts (see Fig. 2): 1) two three-bar linkages (triangles) "bde" and "ghi", which are rigid bodies while the crank rotates; 2) an upper and lower four-bar linkages (crank-rocker) "nm-bj" and "nm-ck", respectively, whose governing equations of motion are similar to any other crank-rocker mechanisms; 3) an open four-bar linkage "dc-fg" called parallel-like linkage because of its figurative resemblance to parallelogram; 4) a rigid three-bar linkage "ghi" called foot; 5) a ground, which is the link between the two pivoted fixed points ($\overline{O_2O_4}$), represented by the dashed vector "n"; and 6) the interconnection point "P" of links "h" and "i", which is called the toe. Each leg is fixed at two points, O_2 and O_4 in Fig. 2. Point O_4 is where the crank is pivoted to the ground and point O_2 is located at the connection of the upper rocker link "b", the lower rocker link "c", and the link "d" of the linkage "bde".

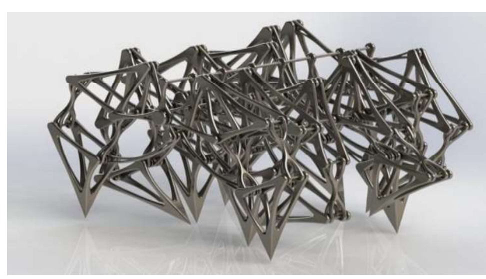


Figure 1. A CAD of the Theo Jansen mechanism [9]

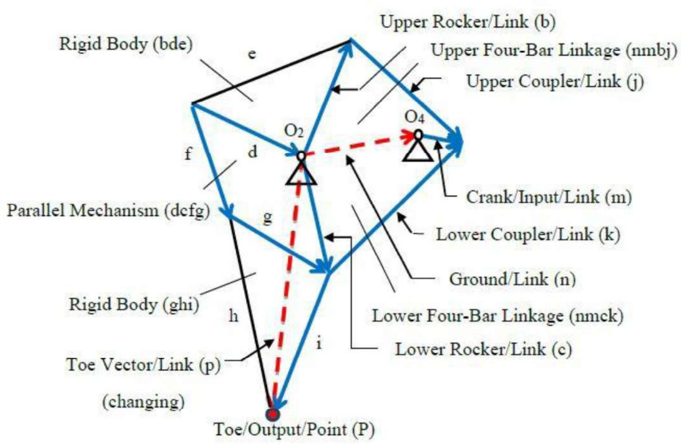


Figure 2. Links and linkages of the Theo Jansen mechanism

Moreover, each two legs are joined together at the point where the crank is fixed to the ground (point O_4), and they are out of phase with each other for one-half cycle of rotation of the crank.

The length of each link in the mechanism is defined to make the foot movement (approximately) linear for one-half of the rotation of the crank. The remaining rotation of the crank allows the foot to raise to a predetermined height, which is called the height of step (see Fig. 3) before returning to the starting position, which is the beginning of the next cycle. During each movement cycle of the Theo Jansen mechanism, each leg experiences four different movement phases which make a closed curve. These phases are called stride phase, lift phase, return phase and lower phase as shown in Fig. 3. The straight line "AB" is called the stride phase in which the leg is in touch with the ground where it should move at a constant velocity. The curve "BCA" consists of two portions, "BC" and "CA". The first portion "BC" is called the lift phase. During this phase the leg moves toward its maximum height. As the leg passes the point "C", it enters into the second portion of the "BCA" which is called the return phase. During the return phase, the leg moves in the same direction as the whole mechanism. The shape of this phase affects the maximum acceleration of the leg. Any attempts to make the curve "CA" as a straight line will ideally minimize the acceleration of the leg. Lastly, the leg descends to the ground until it reaches to the point "A" to complete the movement cycle.

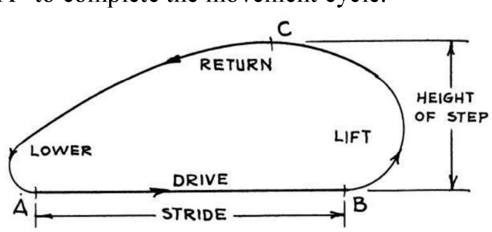


Fig. 3 Movement phases of one leg for each cycle [10]

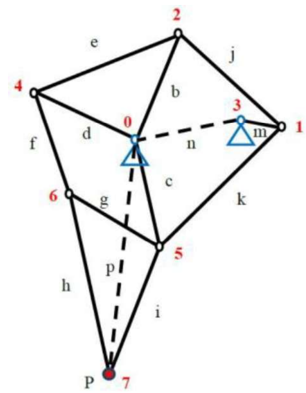


Figure 4. Numbering points for kinematic analysis

KINEMATIC ANALYSIS

In this section we studied the kinematics of one leg of the Theo Jansen mechanism. For this kinematic analysis, we are interested in the position, velocity and acceleration of points 1, 2, 4, 5, 6 and 7 which are shown in Fig. 4. We obtained the general equations of motion for the linkages "nm-bj", "nm-ck", "dc-fg" and "cip" (see Fig. 2) using the loop closure method, in which links are represented as position vectors (see Fig. 5), and then we solved these analytical equations for one full rotation of the crank, by numerical analysis at each degree of rotation.

The upper and lower linkages are categorized as crank-rocker mechanisms because of conforming to the Grashof condition [11]. As in the kinematic analysis we need the dimensions of the links, we used the dimensions that are provided in Theo Jansen's book [12]. To write the equations of motion for each linkage, we used complex numbers and the Euler identity in polar coordinates, which simplifies the calculations.

POSITION ANALYSIS

For position analysis [13] we put the origin of the general coordinate system (GCS) at point O_2 as illustrated in Fig. 5. The vector loop equation for a general four-bar linkage "dc-ab" as shown in Fig. 6 will be:

$$\vec{d} + \vec{c} - \vec{a} - \vec{b} = 0 \qquad (1)$$

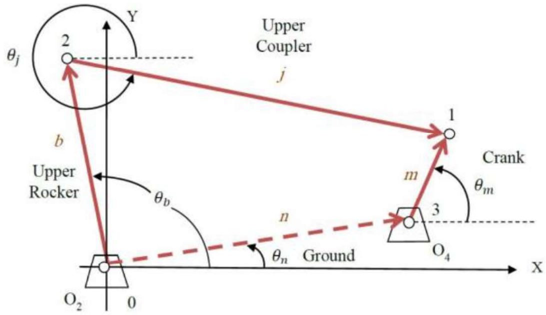
using complex numbers Eq. 1 can be rewritten as:

$$de^{j\theta d} + ce^{j\theta c} - ae^{j\theta a} - be^{j\theta b} = 0 \qquad (2)$$

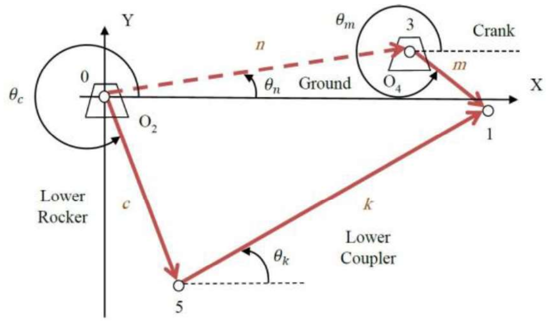
where $\{a, b, c, d\}$ present the link lengths. Equation 2 can be solved for two unknown parameters,the angular positions of the rocker θ_b and coupler θ_a. These unknowns are functions of the links length $\{a, b, c, d\}$ and the angle of the crank θ_c:

$$\theta_{\{a,b\}} = u_{\{a,b\}}(a, b, c, d, \theta_c) \qquad (3)$$

where $\theta_{\{a,b\}}$ means either θ_a or θ_b. As the parameters $\{a, b, c, d\}$ and $\{\theta_a, \theta_b, \theta_c, \theta_d\}$ represent the parameters of the



(a)



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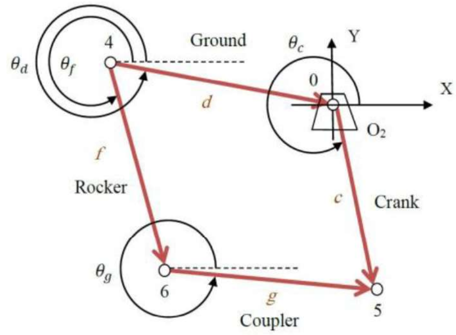
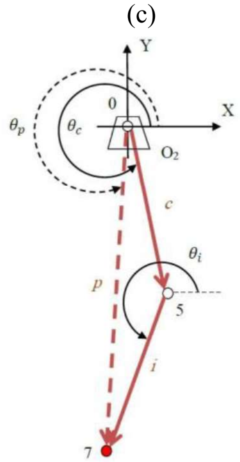


(c)



(d)

Figure 5. Vector loops of the Theo Jansen mechanism;
(a) nm-bj (b) nm-ck (c) dc-fg (d) cip

general linkage "dc-ab", they should be replaced by the corresponding parameters in the linkages under study.

$$Upper: \{a, b, c, d\} \triangleq \{b, j, m, n\} \tag{4}$$

$$\{\theta_a, \theta_b, \theta_c, \theta_d\} \triangleq \{\theta_b, \theta_j, \theta_m, \theta_n\} \tag{5}$$

$$Lower: \{a, b, c, d\} \triangleq \{c, k, m, n\} \tag{6}$$

$$\{\theta_a, \theta_b, \theta_c, \theta_d\} \triangleq \{\theta_c, \theta_k, \theta_m, \theta_n\} \tag{7}$$

$$PARL: \ \{a, b, c, d\} \triangleq \{f, g, c, d\} \tag{8}$$

$$\{\theta_a, \theta_b, \theta_c, \theta_d\} \triangleq \{\theta_f, \theta_g, \theta_c, \theta_d\} \tag{9}$$

for example, Eq. 2 can be written in terms of the real and imaginary parts for the upper linkage:

$$n \cos\theta_n + m \cos\theta_m - b \cos\theta_b - j \cos\theta_j = 0 \tag{10}$$

$$n \sin\theta_n + m \sin\theta_m - b \sin\theta_b - j \sin\theta_j = 0 \tag{11}$$

for the set of Eq. 10 and Eq. 11, the unknowns are $\{\theta_j, \theta_b\}$. From the Eq. 10 and Eq. 11 one can find:

$$x\cos\ \theta_{\{j,b\}} + y\sin\ \theta_{\{j,b\}} \ \triangleq R\cos\left(\theta_{\{j,b\}} - \alpha\right) = A \tag{12}$$

where the constants "x", "y", "R", "α" and "A" are:

$$x = n\cos\theta_n + m\cos\theta_m, \qquad y = n\sin\theta_n + m\sin\theta_m \tag{13}$$

$$R = \sqrt{x^2 + y^2}, \quad \alpha = \tan^{-1}\left(\frac{y}{x}\right) \tag{14}$$

$$A_{\{j,b\}} = \frac{\{j,b\}^2 - \{b,j\}^2 + x^2 + y^2}{2\{j,b\}} \tag{15}$$

In Eq. 12 and Eq. 15, only one subscript inside the curly brackets will be considered at a time. Solving Eq. 12 will determine the angular position of the coupler and rocker of the upper linkage. A similar approach can be used to find the unknown parameters $\{\theta_k, \theta_c\}$ for the lower linkage.

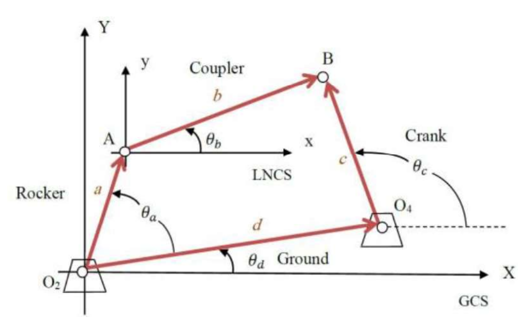
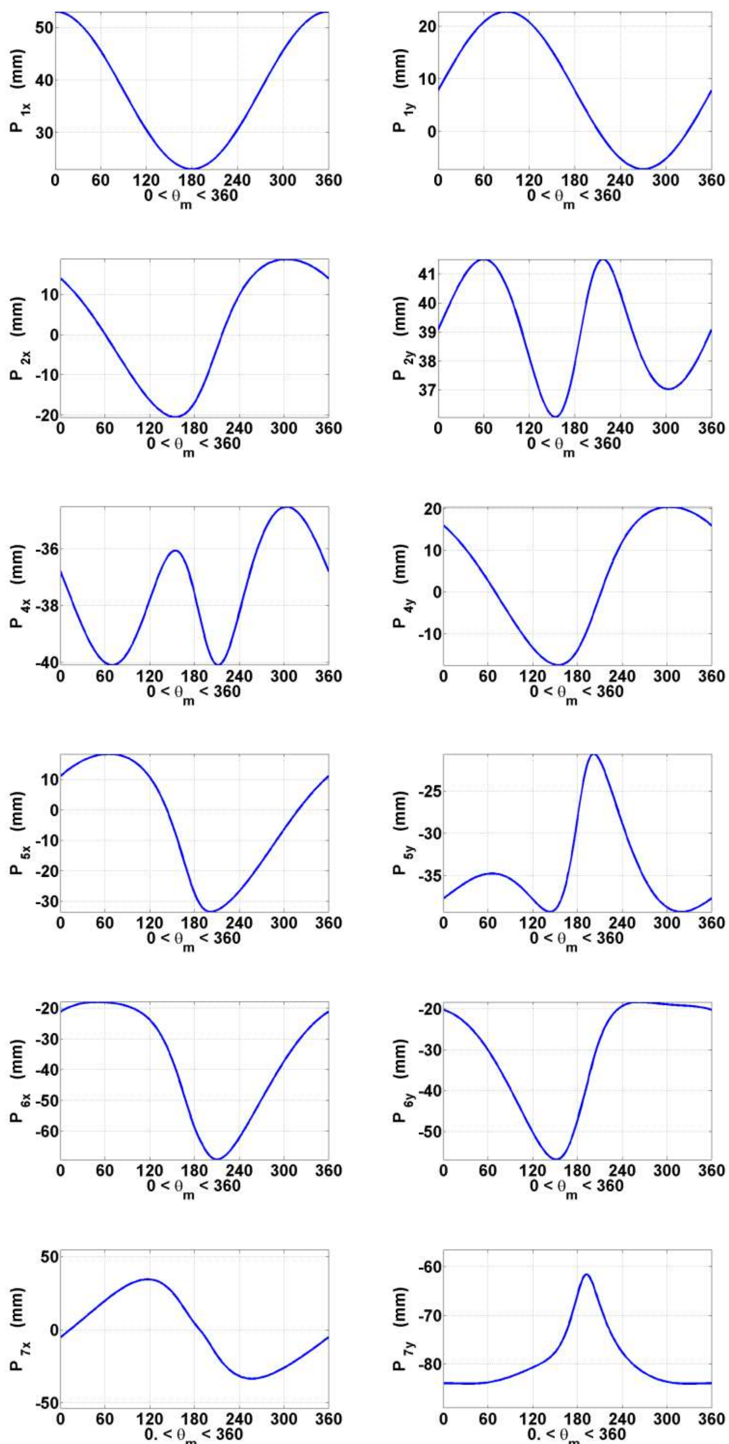


Figure 6. Vector loop of a four-bar linkage



(a) (b)

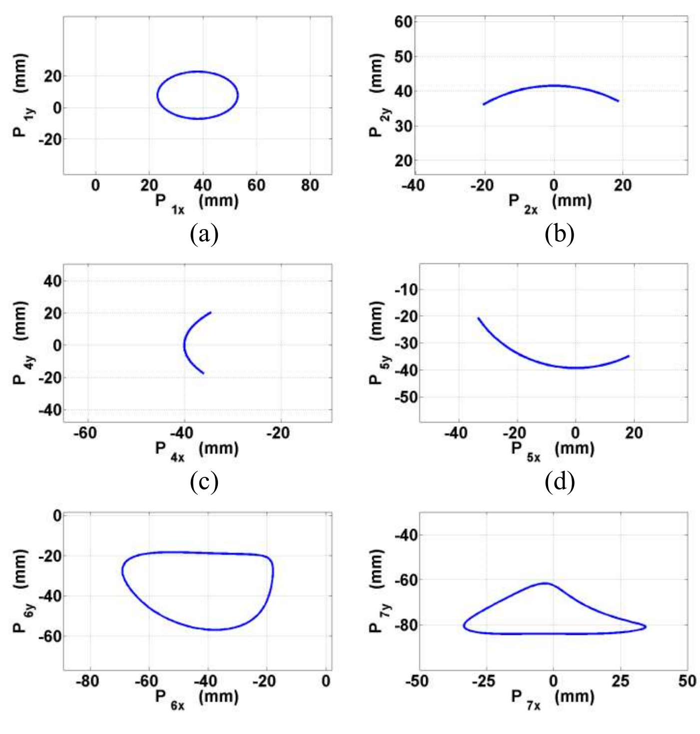
Figure 7. Position of points 1, 2, 4, 5, 6 and 7 for one rotation of the crank; (a) x-components (b) y-components

Since the triangle “bde” has rigid body behavior, the angle of θ_d can be obtained using the Law of Cosines where θ_e is known and θ_b has obtained from solving Eq.12:

$$\theta_d = \theta_e + \theta_b = \cos^{-1}\left(\frac{b^2 + d^2 - e^2}{2bd}\right) + \theta_b \tag{16}$$

The outputs of the kinematic analysis of the upper and lower linkages $\{\theta_c, \theta_d\}$ are used as the inputs of the parallel-like linkage. In the parallel-like linkage “dc-fg”, link “d” is presumed to play the same role as link “n” does in the analysis of the upper and lower linkages. However, its angle is not constant as the crank rotates. Similarly, applying the same method to the parallel-like linkage; separating equations into real and imaginary parts will give unknown parameters of this linkage which are $\{\theta_f, \theta_g\}$.

The angular position θ_g is the foot input. To depict the trace path of the toe (see Fig. 8(f)), it is required to find angular position of the variable dashed vector “p” for each increment of rotation of the crank. This vector together with vectors “c” and “i” creates the triangle “cip” on which the toe is located (see Fig. 5 (d)). The angle θ_i can be found in a similar way as θ_d. The vector loop equation for this triangle gives the angular position of the toe as:



(a) (b) (c) (d) (e) (f)

Figure 8. Loci of all points; (a) Point 1 (b) Point 2 (c) Point 4 (d) Point 5 (e) Point 6 (f) Point 7

$$\theta_p = ln\left(\frac{ce^{j\theta c} + ie^{j\theta i}}{p}\right) \tag{17}$$

The x and y components of the position vector for each point for one full rotation of the crank are illustrated in Fig. 7. Moreover, the locus for each point of interest is shown in Fig. 8.

VELOCITY ANALYSIS

Taking the time derivative of Eq. 2 gives the velocity equation:

$$de^{j\theta d}\frac{d\theta_d}{dt} + ce^{j\theta c}\frac{d\theta_c}{dt} - ae^{j\theta a}\frac{d\theta_a}{dt} - be^{j\theta b}\frac{d\theta_b}{dt} = 0 \tag{18}$$

For the upper and lower linkages, the first and second time derivatives of the ground link "n" are zero. Moreover, the angular velocity of the crank ω_m is assumed to be known and constant at 1 rad/s. Writing the velocity equations in terms of complex numbers for the upper and lower linkages and solving them for unknown parameters $\{\omega_j, \omega_b\}$ and $\{\omega_k, \omega_c\}$ gives:

$$\omega_{\{j,k\}} = \frac{m\omega_m}{\{j,k\}}\frac{sin\left(\theta_m - \theta_{\{b,c\}}\right)}{sin\left(\theta_{\{j,k\}} - \theta_{\{b,c\}}\right)} \tag{19}$$

$$\omega_{\{b,c\}} = \frac{m\omega_m}{\{b,c\}}\frac{sin\ (\theta_m - \theta_{\{j,k\}})}{sin\ (\theta_{\{b,c\}} - \theta_{\{j,k\}})} \tag{20}$$

Correspondingly, the linear velocity for each link will be:

$$V_m = m\omega_m(-sin\theta_m + jcos\theta_m) \tag{21}$$

$$V_{\{j,k\}} = \{\mathrm{j},k\}\omega_{\{j,k\}}(-sin\theta_{\{j,k\}} + jcos\theta_{\{j,k\}}) \tag{22}$$

$$V_{\{b,c\}} = \{b,c\}\omega_{\{b,c\}}(-sin\theta_{\{b,c\}} + jcos\theta_{\{b,c\}}) \tag{23}$$

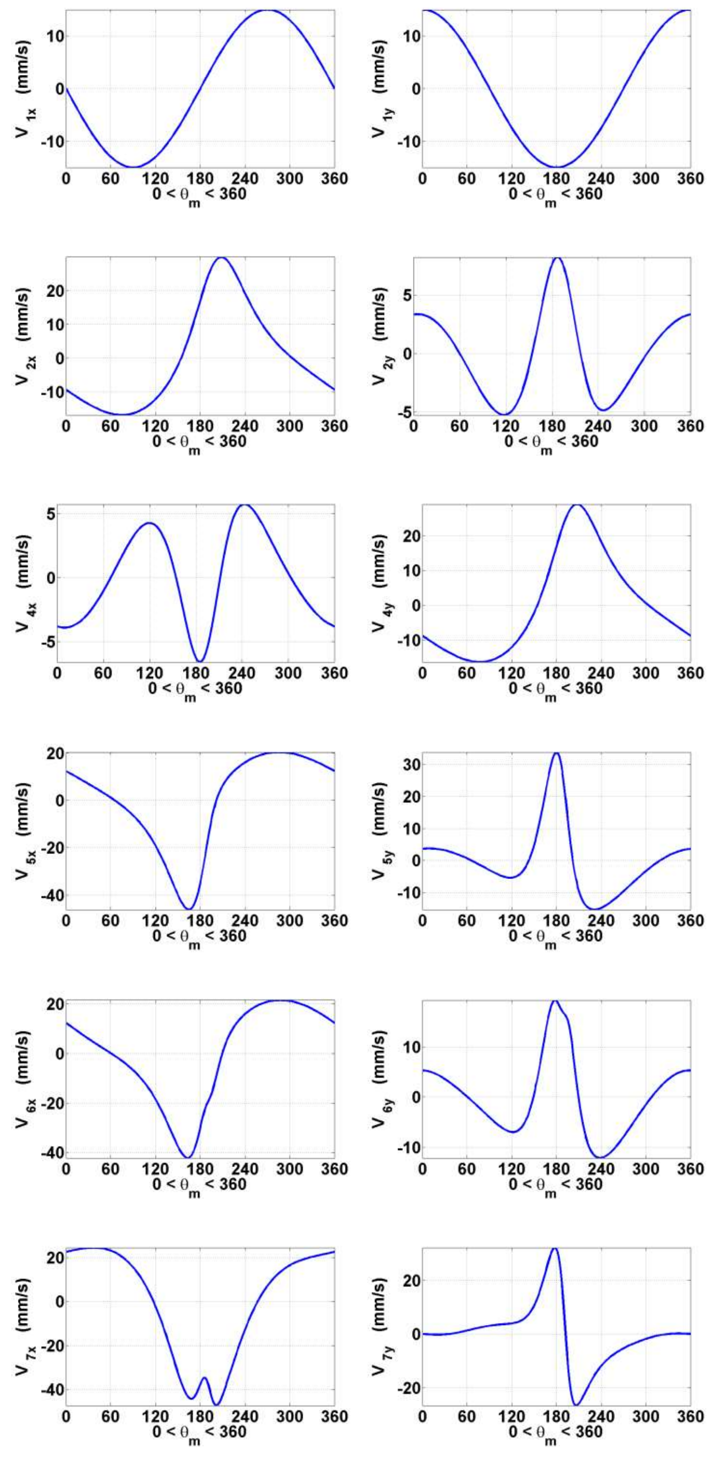
To find vectors of angular velocities for the parallel-like linkage, Eq. 18 can be written in the form of known and unknown matrices:

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix}\begin{bmatrix} \omega_f \\ \omega_g \end{bmatrix} = \begin{bmatrix} R_1 \\ R_2 \end{bmatrix} \tag{24}$$

where:

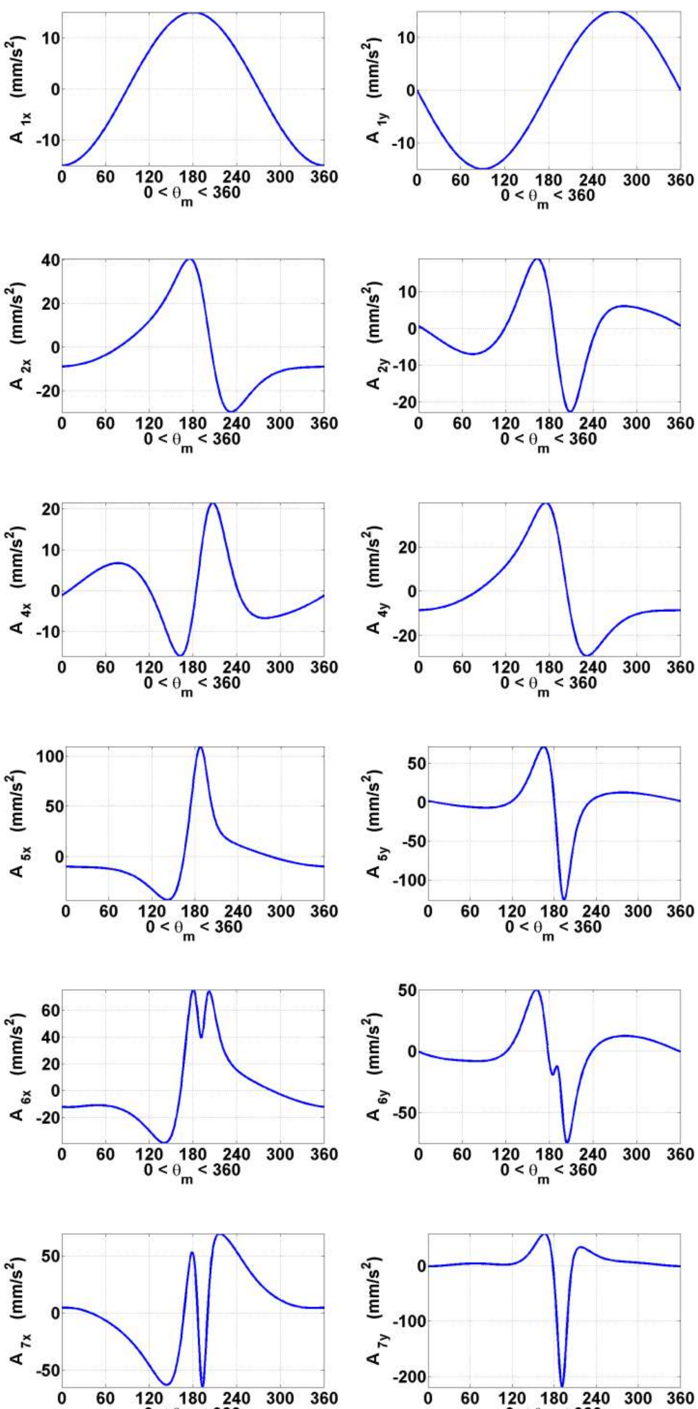
$$\begin{aligned} A_1 &= f\,sin\,\theta_f \\ A_2 &= g\,sin\,\theta_g \\ A_3 &= f\,cos\,\theta_f\,, \\ A_4 &= g\,cos\,\theta_g \end{aligned} \quad \begin{aligned} R_1 &= d\omega_d\,sin\,\theta_d + c\omega_c\,sin\,\theta_c \\ R_2 &= d\omega_d\,cos\,\theta_d + c\omega_c\,cos\,\theta_c \end{aligned} \tag{25}$$

As mentioned before, the triangle "bde" acts kinematically as a rigid body, hence $\omega_d = \omega_b$. Similar rigid body behavior is also obtained for the linkage "ghi" where $\omega_p = \omega_i = \omega_g$. The components of linear velocity of the toe can be written as



(a) (b)

Figure 9. Linear velocities of points 1, 2, 4, 5, 6 and 7 for one rotation of the crank; (a) x-components (b) y-components



(a) (b)

Figure 10. Linear accelerations of points 1, 2, 4, 5, 6 and 7 for one rotation of the crank; (a) x-components (b) y-components

the summation of the x and y components of the linear velocities of the link "c" and link "i" (see Fig. 5(d)):

$$V_{p_x} = V_{c_x} + V_{i_x}, \qquad V_{p_y} = V_{c_y} + V_{i_y} \tag{26}$$

The x and y components of the linear velocity for all points (see Fig. 4) for one full rotation of the crank are shown in Fig. 9.

ACCELERATION ANALYSIS

Acceleration equations are derived from differentiating of the velocity equations. Simplifying along with grouping terms gives:

$$\begin{aligned} &ae^{j\theta_a}(-\omega_a{}^2 + \alpha_a) + \\ &be^{j\theta_b}(-\omega_b{}^2 + \alpha_b) - \\ &ce^{j\theta_c}(-\omega_c{}^2 + \alpha_c) = 0 \end{aligned} \tag{27}$$

All the links lengths, angles, and angular velocities have been determined so far. If the input angular acceleration of the crank is known, then the angular acceleration of the coupler and rocker of the upper and lower linkages will be functions of only aforementioned parameters:

$$\alpha_b = w_b(a, b, c, d, \theta_a, \theta_b, \theta_c, \omega_a, \omega_b, \omega_c, \alpha_a) \tag{28}$$

$$\alpha_c = v_c(a, b, c, d, \theta_a, \theta_b, \theta_c, \omega_a, \omega_b, \omega_c, \alpha_a) \tag{29}$$

Expanding the Eulerian form of Eq. 27 in terms of complex numbers; rearranging real and imaginary parts solely to solve for angular accelerations of the upper and lower linkages leads to:

$$\begin{aligned} \alpha_{\{j,k\}} &= \frac{CD - AF}{AE - BD} \\ \alpha_{\{b,c\}} &= \frac{CE - BF}{AE - BD} \end{aligned} \tag{30}$$

where:

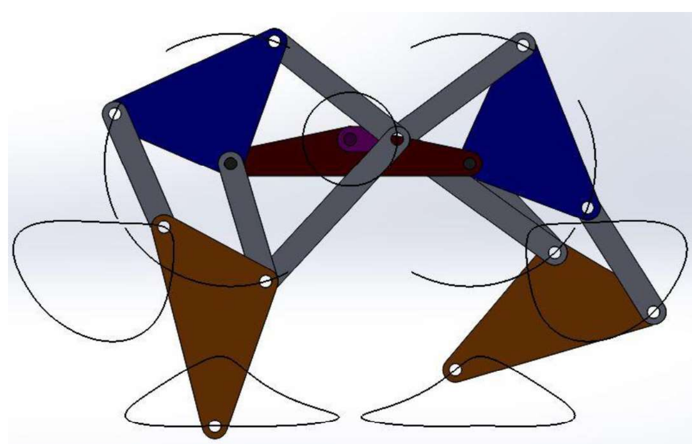


Figure 11. Trace path (loci) of all points of a paired leg using SolidWorks Motion Study Package

$$A = \{b, c\}sin\theta_{\{b,c\}} \tag{31}$$

$$B = \{j, k\}sin\theta_{\{j,k\}} \tag{32}$$

$$\begin{aligned} C = m\alpha_m sin\theta_m + m{\omega_m}^2 cos\theta_m \\ +\{j, k\}{\omega_{\{j,k\}}}^2 cos\theta_{\{j,k\}} \\ -\{b, c\}{\omega_{\{b,c\}}}^2 cos\theta_{\{b,c\}} \end{aligned} \tag{33}$$

$$D = \{b, c\}cos\theta_{\{b,c\}} \tag{34}$$

$$E = \{j, k\}cos\theta_{\{j,k\}} \tag{35}$$

$$\begin{aligned} F = m\alpha_m cos\theta_m - m{\omega_m}^2 sin\theta_m \\ -\{j, k\}{\omega_{\{j,k\}}}^2 sin\theta_{\{j,k\}} \\ +\{b, c\}{\omega_{\{b,c\}}}^2 sin\theta_{\{b,c\}} \end{aligned} \tag{36}$$

Now the components of linear acceleration for each link will be:

$$\begin{aligned} A_m &= m\alpha_m(-sin\theta_m + jcos\theta_m) \\ &\quad - m{\omega_m}^2(cos\theta_m + jsin\theta_m) \\ A_{m_x} &= -m\alpha_m sin\theta_m - m{\omega_m}^2 cos\theta_m \\ A_{m_y} &= m\alpha_m cos\theta_m - m{\omega_m}^2 sin\theta_m \end{aligned} \tag{37}$$

$$\begin{aligned} A_{\{j,k\}} &= \{j, k\}\alpha_{\{j,k\}}\left(-sin\theta_{\{j,k\}} + jcos\theta_{\{j,k\}}\right) \\ &\quad - \{j, k\}{\omega_{\{j,k\}}}^2\left(cos\theta_{\{j,k\}} \right. \\ &\quad \left. + jsin\theta_{\{j,k\}}\right) \\ A_{\{j,k\}_x} &= -\{j, k\}\alpha_{\{j,k\}} sin\theta_{\{j,k\}} \\ &\quad - \{j, k\}{\omega_{\{j,k\}}}^2 cos\theta_{\{j,k\}} \\ A_{\{j,k\}_y} &= \{j, k\}\alpha_{\{j,k\}} cos\theta_{\{j,k\}} - \{j, k\}{\omega_{\{j,k\}}}^2 sin\theta_{\{j,k\}} \end{aligned} \tag{38}$$

$$\begin{aligned} A_{\{b,c\}} &= \{b, c\}\alpha_{\{b,c\}}\left(-sin\theta_{\{b,c\}} + jcos\theta_{\{b,c\}}\right) \\ &\quad - \{b, c\}{\omega_{\{b,c\}}}^2\left(cos\theta_{\{b,c\}} \right. \\ &\quad \left. + jsin\theta_{\{b,c\}}\right) \\ A_{\{b,c\}_x} &= -\{b, c\}\alpha_{\{b,c\}} sin\theta_{\{b,c\}} \\ &\quad - \{b, c\}{\omega_{\{b,c\}}}^2 cos\theta_{\{b,c\}} \\ A_{\{b,c\}_y} &= \{b, c\}\alpha_{\{b,c\}} \cos\theta_{\{b,c\}} - \{b, c\}{\omega_{\{b,c\}}}^2 \sin\theta_{\{b,c\}} \end{aligned} \tag{39}$$

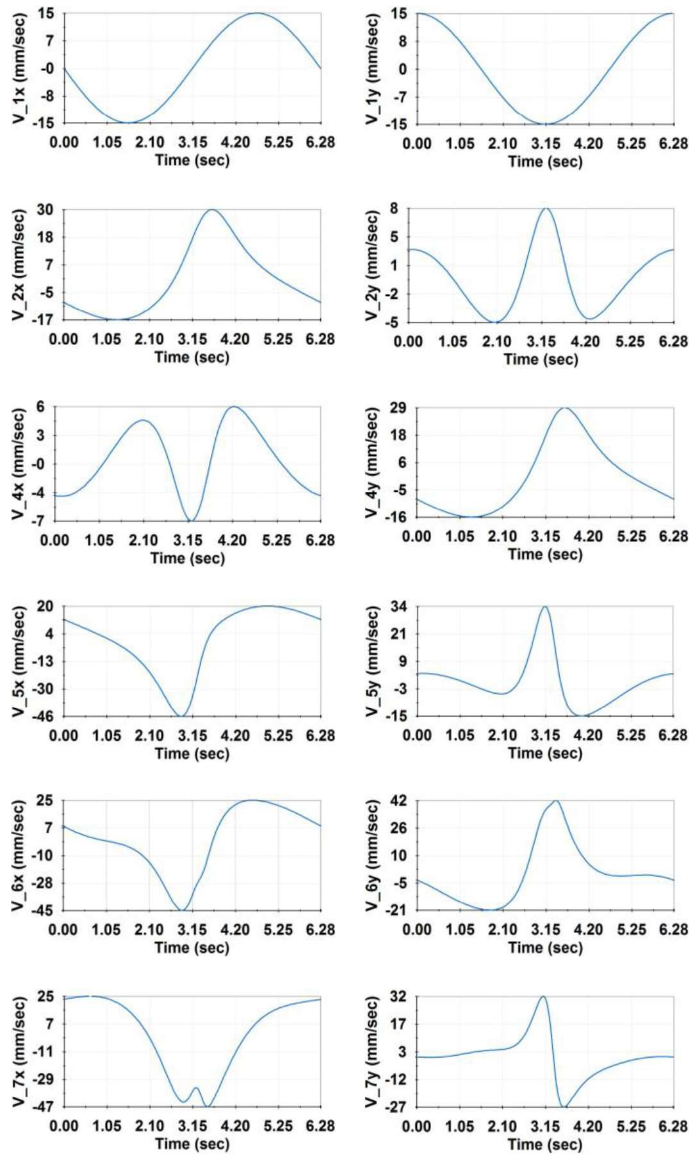
We also can employ the same method used in the velocity analysis section to write the acceleration equations for the parallel-like linkage and solve them for the unknown parameters $\{\alpha_f, \alpha_g\}$:

$$\begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} \begin{bmatrix} \alpha_f \\ \alpha_g \end{bmatrix} = \begin{bmatrix} C_1 \\ C_2 \end{bmatrix} \tag{40}$$

where:

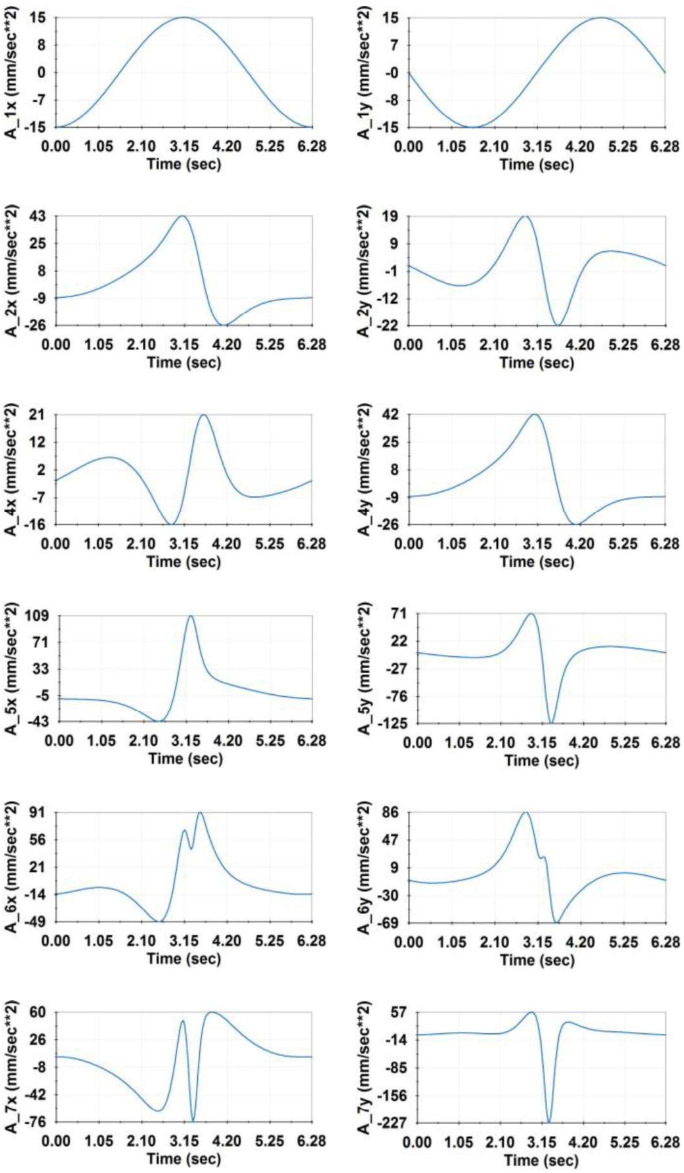
$$\begin{aligned} B_1 &= -f\ sin\ \theta_f \\ B_2 &= -g\ sin\ \theta_g \\ B_3 &= f\ cos\ \theta_f \\ B_4 &= g\ cos\ \theta_g \end{aligned} \tag{41}$$

$$\begin{aligned} C_1 &= -d{\omega^2}_d\, cos\, \theta_d - d\alpha_d\, sin\, \theta_d - c{\omega^2}_c\, cos\, \theta_c \\ &\quad -c\alpha_c\, sin\, \theta_c + f{\omega^2}_f\, cos\, \theta_f + g{\omega^2}_g\, cos\, \theta_g \\ C_2 &= -d{\omega^2}_d\, sin\, \theta_d + d\alpha_d\, cos\, \theta_d - c{\omega^2}_c\, sin\, \theta_c \\ &\quad +c\alpha_c\, cos\, \theta_c + f{\omega^2}_f\, sin\, \theta_f + g{\omega^2}_g\, sin\, \theta_g \end{aligned} \tag{42}$$



(a) (b)

Figure 12. Linear velocities of points 1, 2, 4, 5, 6 and 7 for the equivalent time for one rotation of the crank using SolidWorks Motion Study Package;
(a) x-components (b) y-components



(a) (b)

Figure 13. Linear accelerations of points 1, 2, 4, 5, 6 and 7 for the equivalent time for one rotation of the crank using SolidWorks Motion Study Package;
(a) x-components (b) y-components

Since the triangle "bde" moves rigidly, the angular accelerations of all its links will be identical, meaning $\alpha_d = \alpha_b$.

Likewise, for the foot linkage "ghi", we have $\alpha_i = \alpha_p = \alpha_g$. Then linear acceleration of the point "P" can be simply written as (see Fig. 5 (d)):

$$A_{p_x} = A_{c_x} + A_{i_x}, \qquad A_{p_y} = A_{c_y} + A_{i_y} \tag{43}$$

The x and y components of the linear acceleration for all the points (see Fig. 4) for one full rotation of the crank are shown in Fig. 10.

SOLIDWORKS SIMULATION

In order to measure the precision of the analytical results, we simulated the motion of the Theo Jansen mechanism using SolidWorks Motion Study package. The position (loci), velocity and acceleration results for the points 1, 2, 4, 5, 6 and 7 are represented in Fig. 11, Fig. 12 and Fig. 13, respectively. The linear velocities and the linear accelerations plots obtained from SolidWorks are shown for the time of 6.28 seconds which is equal to one full rotation of the crank at constant angular velocity of 1 rad/s.

CONCLUSIONS

In this paper we employed the loop closure method for kinematic analysis of the Theo Jansen mechanism. Each leg of this mechanism is divided into four linkages, which are called upper, lower, parallel-like and "cip" linkages. Furthermore, the kinematics is studied for six points of the leg for one full rotation of the crank, using MATLAB. A simulation was performed to evaluate the analytical results for each point of interest using SolidWorks Motion Study package for the equivalent time for one full rotation of the crank. The position (loci), linear velocity and linear acceleration results for the desired points obtained in SolidWorks were in good agreement with those obtained from the analytical approach in MATLAB.

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